

ATPL EASA STUDY GUIDE | 050 - METEOROLOGY

Chapter 10: Pressure Systems: Depressions, Anticyclones & Col

Syllabus Source: CAE Oxford (pp. 323-358) | Pass Target: >= 90% AviationExam

1. LOW PRESSURE SYSTEMS (DEPRESSIONS / CYCLONES)

A depression is an area of low barometric pressure surrounded by closed isobars. In the Northern Hemisphere, wind circulates COUNTER-CLOCKWISE and inwards towards the center:

- Polar Front Depressions: Form along the boundary between cold polar and warm tropical air. Life cycle: Initial wave -> Developing depression -> Mature occlusion -> Decaying low.
- Thermal Depressions (Heat Lows): Form over hot continental land masses in summer due to intense solar heating (e.g. Iberian Peninsula, Persian Gulf). Stationary, shallow, dissipating at night.
- Orographic Depressions (Lee Lows): Form on the leeward side of mountain ranges as upper air crosses the ridge (e.g. Genoa Low south of the Alps).

2. HIGH PRESSURE SYSTEMS (ANTICYCLONES)

An anticyclone is an area of high barometric pressure with widely spaced isobars. Wind circulates CLOCKWISE and outwards in the Northern Hemisphere:

- Characteristics of Anticyclones: SUBSIDENCE: Air gently sinks from high levels, warming adiabatically. Sinking air suppresses vertical convection, producing stable, cloudless skies at upper levels.
- Subtropical Anticyclones: Permanent warm-core highs (e.g. Azores High) driving global trade winds.
- Cold Continental Anticyclones: Winter highs over snow-covered land (Siberian High, Canadian High). Extremely cold, dense surface air with intense ground inversions.
- Blocking Highs: Large, stagnant high-pressure cells that deflect mid-latitude depressions for weeks, causing persistent weather patterns.

3. RIDGES, TROUGHS & COLS

- Ridge of High Pressure: An elongated wedge of high pressure extending from an anticyclone. Brings improved, settled weather.
- Trough of Low Pressure: An elongated area of low pressure. Acts like a cold front, producing cloud lines, squalls, and showers.
- Col: The neutral saddle-point region between two highs and two lows. Characterized by calm or very light winds. Highly prone to RADIATION FOG IN WINTER, and SEVERE THUNDERSTORMS IN SUMMER!

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Anticyclone Types & Col Recognition

Anticyclone Characteristics: - Clockwise circulation (Northern Hemisphere), surface divergence, central subsidence (sinking air). - Types: Subtropical Highs (warm, deep, e.g. Azores); Thermal/Continental Highs (shallow, cold, e.g. Siberian winter high); Blocking Highs (diverts storm track for weeks). - Col: Neutral saddle region between two highs and two lows; calm winds, unpredictable weather (fog in winter, thunderstorms in summer).

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)**Q1: What weather conditions are typically associated with a 'COL' during summer afternoons over land?**

- [A]: Gale-force winds
- [B]: Light winds and heavy SEVERE THUNDERSTORMS caused by daytime surface heating
- [C]: Persistent radiation fog
- [D]: Snow flurries

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. In summer, the calm winds and stagnation of a col allow strong diurnal surface heating to trigger violent convective thunderstorms.

Q2: In an ANTICYCLONE, why is cloud formation generally suppressed at middle and high altitudes?

- [A]: Strong winds blow clouds away
- [B]: Because of AIR SUBSIDENCE (sinking air warms adiabatically, lowering relative humidity)
- [C]: Air is too cold to form clouds
- [D]: Pressure is too low

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Sinking air warms at DALR (3°C/1,000 ft), moving further away from saturation (lower relative humidity) and evaporating cloud droplets.

Q3: What is a 'BLOCKING HIGH' in European meteorology?

- [A]: A small thunderstorm cell
- [B]: A large, quasi-stationary anticyclone that remains in place for days or weeks, blocking and deflecting approaching Atlantic depressions
- [C]: An inversion at 1,000 ft
- [D]: A hurricane eye

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Blocking anticyclones divert the mid-latitude westerly storm track to the north or south, locking the region into persistent dry or cold weather.